JUNIOR SCHOOL LEARNING HUB

GRADE 9 INTEGRATED SCIENCE
END-OF-TERM 3 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 3	Learning Area: Integrated Science

INSTRUCTIONS TO LEARNERS

- Answer all questions in Section A and Section B.
- Write clear scientific explanations and show calculations where required.
- Use labelled diagrams where appropriate.
- Observe all laboratory and environmental safety precautions.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Define the term ecosystem. (2 marks)

2. State two biotic components of an ecosystem. (2 marks)

3. Give two examples of abiotic factors in an ecosystem. (2 marks)

4. What is a food chain? (2 marks)

5. State two ways in which human activities can disrupt food chains. (2 marks)

6. Name two methods of conserving biodiversity. (2 marks)

7. State two functions of the human reproductive system. (2 marks)

8. Name the male reproductive cell and the female reproductive cell in humans. (2 marks)

9. State two physical changes that may occur during puberty. (2 marks)

10. Give two ways of maintaining personal hygiene during adolescence. (2 marks)

11. Define the term communicable disease. (2 marks)

12. State two ways communicable diseases can spread. (2 marks)

13. Give two measures used to prevent the spread of infectious diseases. (2 marks)

14. Name two non-communicable diseases. (2 marks)

15. State two effects of drug and substance abuse on young people. (2 marks)

16. Define the term force. (2 marks)

17. State the SI unit of force. (2 marks)

18. A force of 20 N acts on a mass of 5 kg. Calculate the acceleration. (2 marks)

19. State two ways of reducing friction where it is undesirable. (2 marks)

20. Give two examples of energy transformations in everyday life. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Ecosystems and conservation (10 marks)

a) Distinguish between a food chain and a food web. (2 marks)

b) Construct a food chain using grass, a grasshopper, a frog and a snake. (2 marks)

c) Explain three ways human activities can affect the balance of an ecosystem. (3 marks)

d) State three actions communities can take to conserve biodiversity. (3 marks)

22. Human reproduction and adolescence (10 marks)

a) Name four organs of the human female reproductive system. (4 marks)

b) State two functions of the testes. (2 marks)

c) Explain two ways adolescents can maintain physical and emotional wellbeing. (2 marks)

d) State two responsible practices that help prevent sexually transmitted infections. (2 marks)

23. Health, diseases and substance abuse (10 marks)

a) Distinguish between communicable and non-communicable diseases. (2 marks)

b) State three ways infectious diseases can be transmitted. (3 marks)

c) Explain three measures that reduce the spread of infectious diseases in a community. (3 marks)

d) State two ways drug and substance abuse can affect learning. (2 marks)

24. Forces and motion (10 marks)

a) State Newton's second law of motion. (2 marks)

b) A net force of 36 N acts on a 9 kg object. Calculate its acceleration. (3 marks)

c) Explain two effects a force may have on an object. (2 marks)

d) State three factors that affect friction between two surfaces. (3 marks)

25. Energy, work and simple machines (10 marks)

a) Define work done in science. (2 marks)

b) A learner pushes a box with a force of 50 N through 4 m in the direction of the force. Calculate the work done. (3 marks)

c) State three ways simple machines make work easier. (3 marks)

d) Give two examples of simple machines used at home or school. (2 marks)

26. Scientific investigation and environmental health (10 marks)

a) State four steps followed in a scientific investigation. (4 marks)

b) Explain why a fair test changes only one independent variable at a time. (2 marks)

c) State two ways of managing solid waste responsibly. (2 marks)

d) Give two reasons why clean water and proper sanitation are important to health. (2 marks)

MARKING SCHEME

Grade 9 Integrated Science – End-of-Term 3 Examination

Total: 100 marks | Section A: 40 marks | Section B: 60 marks

SECTION A (40 MARKS)

1. A community of living organisms interacting with one another and with the non-living environment. (2)

2. Any two: plants, animals, fungi, bacteria and other microorganisms. (2)

3. Any two: light, temperature, water, air, soil, minerals, humidity. (2)

4. A sequence showing the transfer of energy as one organism is eaten by another. (2)

5. Any two: habitat destruction, pollution, overhunting/overfishing, pesticide use, introduction of invasive species. (2)

6. Any two: habitat protection, tree planting, wildlife protection, seed banks, sustainable resource use, control of pollution. (2)

7. Any two: production of gametes; fertilisation; support of development of embryo/foetus; childbirth; hormone production. (2)

8. Sperm cell and ovum/egg cell. (2)

9. Any two: growth spurt, body hair growth, voice changes, breast development, menstruation, changes in body shape, skin changes. (2)

10. Any two: regular bathing; clean clothing; oral hygiene; handwashing; appropriate menstrual hygiene; keeping private body areas clean. (2)

11. A disease caused by an infectious agent that can be transmitted directly or indirectly between hosts. (2)

12. Any two: droplets/air; contaminated food or water; direct contact; vectors; blood/body fluids; contaminated objects. (2)

13. Any two: vaccination, handwashing, sanitation, safe food/water, ventilation, vector control, respiratory hygiene. (2)

14. Any two: diabetes, cancer, hypertension, asthma, cardiovascular disease. (2)

15. Any two: poor school performance, dependence, illness, accidents, mental health problems, damaged relationships. (2)

16. A push or pull that can change or tend to change an object's state of motion or shape. (2)

17. Newton (N). (2)

18. $a = F/m = 20/5 = 4\ m/s^2$. (2)

19. Any two: lubrication; use rollers/ball bearings; smooth surfaces; streamline objects where relevant. (2)

20. Any two: electrical to light (lamp); chemical to electrical (battery); electrical to sound (speaker); chemical to kinetic/heat (fuel); potential to kinetic (falling object). (2)

SECTION B (60 MARKS)

21. Ecosystems and conservation (10 marks)

A food chain shows one pathway of energy transfer; a food web shows interconnected food chains in an ecosystem. (2)

Grass → grasshopper → frog → snake. (2)

Deforestation destroys habitats; pollution harms organisms; overfishing/overhunting reduces populations; use of pesticides affects non-target organisms; invasive species disrupt interactions. Any three explained. (3)

Protect habitats; plant indigenous trees; prevent poaching; reduce pollution; establish protected areas; practise sustainable resource use; control invasive species. Any three. (3)

22. Human reproduction and adolescence (10 marks)

Ovaries, fallopian tubes/oviducts, uterus, cervix, vagina. Any four. (4)

Produce sperm cells and secrete testosterone. (2)

Balanced diet, exercise, adequate sleep, personal hygiene, seeking trusted guidance, positive friendships, managing stress. Any two explained. (2)

Abstinence; avoiding risky sexual behaviour; seeking accurate health information and appropriate health services; mutual faithfulness in a mutually faithful relationship; correct condom use as public-health guidance. Any two. (2)

23. Health, diseases and substance abuse (10 marks)

Communicable diseases can spread from one person/organism to another; non-communicable diseases do not spread by infection between people. (2)

Airborne droplets; contaminated food or water; direct contact; contaminated objects; vectors; blood/body fluids. Any three. (3)

Handwashing; vaccination; safe water and food; sanitation; ventilation; respiratory hygiene; vector control; early diagnosis and appropriate treatment. Any three explained. (3)

Poor concentration; memory problems; absenteeism; reduced performance; risky behaviour; dependence and health problems. Any two. (2)

24. Forces and motion (10 marks)

The resultant force on an object is equal to the rate of change of its momentum; for constant mass, $F = ma$. (2)

$F = ma$; $a = F/m = 36/9 = 4\ m/s^2$. Formula (1), substitution (1), answer and unit (1). (3)

Change speed; change direction; change shape; start or stop motion. Any two. (2)

Nature/roughness of surfaces; force pressing surfaces together; lubrication; type of materials. Any three relevant factors. (3)

25. Energy, work and simple machines (10 marks)

Work is done when a force causes displacement in the direction of the force; W = F × d for a constant force in that direction. (2)

W = Fd = 50 × 4 = 200 J. Formula (1), substitution (1), answer/unit (1). (3)

Multiply force; change direction of force; increase speed/distance moved; allow force to be applied more conveniently. Any three. (3)

Lever, pulley, inclined plane, wheel and axle, wedge, screw. Any two. (2)

26. Scientific investigation and environmental health (10 marks)

Identify a problem/question; formulate hypothesis; plan investigation; collect data; analyse results; draw conclusion; communicate findings. Any four. (4)

It helps establish whether changes in the dependent variable are due to the variable being tested, while other factors are controlled. (2)

Reduce, reuse, recycle, compost biodegradable waste, segregate waste, use approved disposal facilities. Any two. (2)

Prevent waterborne diseases; maintain hygiene; protect water sources; improve community wellbeing. Any two. (2)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Integrated Science teacher should verify that the assessed concepts match the actual Grade 9 Term 3 teaching sequence and the school scheme of work. Accept scientifically correct equivalent responses and award method marks where indicated.

Prepared by: Junior School Learning Hub